Final-Progress Report

Replicating General Circulation Models with Cellular Automata

By Aditi Iyer
Faculty Advisor: Dr N. Sukumar
School of Natural Sciences, Department of Chemistry
25th July 2022

Table of Contents -

Abstract 0
Introduction 1
Methodology 2
Results and Discussion 7
References 9
Acknowledgements 9
Reflection on the Learning Experience - 8
Appendix 10

Abstract

Cellular Automata have been used to model various physical phenomena, including but not limited to, the Navier-Stokes equation, Sedimentation and Percolation. However, they are yet to be employed in modelling climate systems. Cellular Automata are used as alternatives to differential equations thus making them a potential contender for generating Climate Models. In the following report, we have tried to model a basic geodesic polyhedron using spherical triangles, to model the Earth. Each face of the polyhedron is taken to be a cell and the Cellular Automaton applies the rules to them. The rules for interaction are based on the Navier Stokes equation.

Introduction

Climate models play a crucial role in policy and in understanding our role in the future of the planetary ecosystem. Many highly sophisticated climate models have been made based on various types of machine learning programs.

Cellular Automata (CA) are computational models that simulate complex systems[1]. The idea is to create a lattice of cells. Every cell in the system has a defined state in the beginning. The cells interact with neighbouring cells with which they share an edge, and these interactions are governed by a set of predefined rules. The automaton is run for several time steps, where at each step, the new state of any cell is formed by its interaction with its neighbours from the previous time step according to the Automaton's rule.

Datasets for Temperature were used from Berkely Earth[2], which is an independent U.S. non-profit organisation focused on environmental data science. Other data sets will be used to cross-reference the results obtained from different sources. Similar Data-sets for Pressure were used from the NCAR's Data Archive[3].

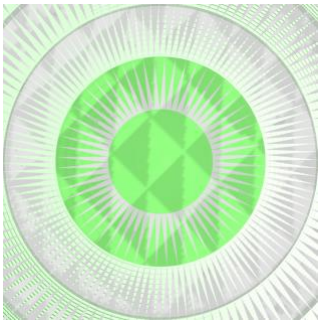
In the upcoming versions, we will try to run time steps that include pressure terms as well as the initial state of the automaton.

The model assumes the Earth to be an exact sphere. The sphere is divided into 120 longitudes and 60 latitudes (30 North and 30 South, conventionally), making 7200 cells. Each parallelogram (further detailed geometry discussed below) formed by the latitudes and longitudes is split into two triangles, one upper and one lower. This gives us a total of 15400 cells. Each cell is then assigned an initial temperature. The temperature is an average over a definite volume in a given time period. Temperatures used to describe the initial states of the automaton are averages taken over latitude and longitude. As the automaton is discrete with respect to both space and time components, the model atmosphere becomes a discrete dynamical system.

Methodology and Results

The most important element of any climate model is the Navier-Stokes equation. The equation is used to describe the atmosphere. But the obvious problem remains that the Navier-stokes equation does not have an analytical solution. Coupled with this, applying the Navier-Stokes equation to a three-dimensional system is impossible. Thus, on a three-dimensional lattice, it is solved numerically. To avoid the problem entirely, we used a two-dimensional geodesic sphere to apply the Navier-Stokes equation to the lattice without having to compute it numerically for each point. We have applied a simple additive temperature rule to the cellular automata, for a two-dimensional lattice.

The lattice is formed out of parallelograms formed out of latitudes and longitudes, which are further dissected into triangles using diagonals. They are not exact parallelograms as the lattice is warped into 3 dimensions, so the edges are curved making the east and west sides such that the longitudes meet at the pole. But for ease of discussion, they will be referred to as parallelograms as their nearest 2D counterparts are parallelograms. The first band of parallelograms eliminates the upper triangles, and gives all the lower triangles sharing a vertex, thus forming the top of the sphere. The same is done for the last band of triangles, by eliminating the lower triangles. Fig 1.1 is a close-up of the upper and lower bands on triangles.



Thus the lattice mimics a sphere and can be considered a simpler version of the Earth.

(Fig 1.1 First band of triangles shares one vertex, the green circle in the center is the vertice shared by the first band of triangles. The grey square grid is the visualizing software's gird)

I took the temperature data from Berkely Earth[2], where the original data set spans from 1853 to 2013 and 36.17°S to 50.63°N and 175.03°W to 16.48°E, but I only used the data from 1990 onwards, as the source of the earlier data set is still unclear. I am still looking for alternate data sets that can be used to run the same calculation to verify if the calculation is consistent across various data sets. I have been unable to find alternate openly available data sets but have approached Copernicus and NOAA for the same.

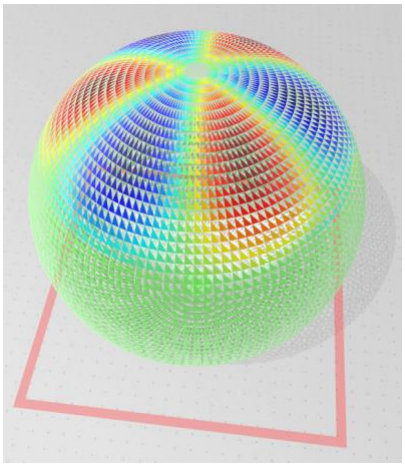
The data set was then sorted by month and five separate months were chosen at roughly equal intervals. These were January, April, June, August and November. Then the temperature versus Latitude was calculated for the years 1990, 1995, 2000, 2005 and 2010. The average gradient was calculated across the years and this equation was used to scale the temperatures according to the longitudes and latitudes of our 120 by 60 lattice.

The gradient from north to south was, (80.6 + (0.374*x) - (0.0121*(x^2))).

The gradient from east to west was, (0.064164*x + 57.54)

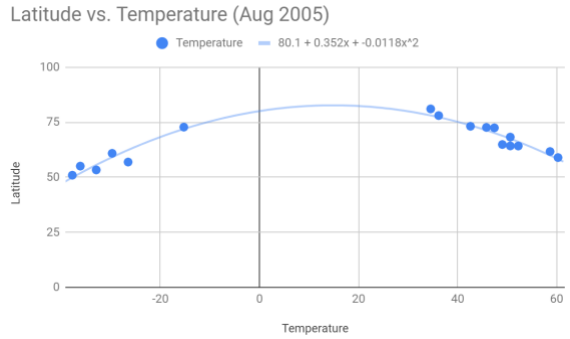
The above equations show that the gradient along the north-south direction, with respect to the number of cells, is significantly higher than in the east-west direction.

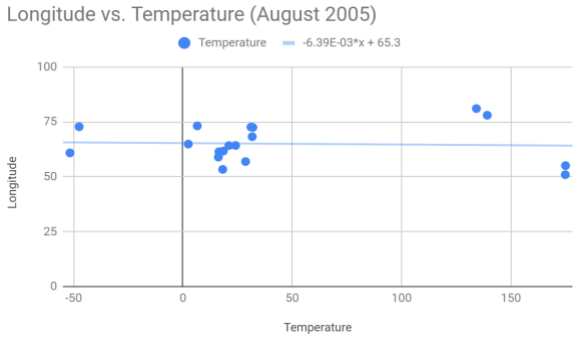
In the first run, I did not add the gradients for longitudes or latitudes and input random values for the initial states and I got the following result (Fig 1.2);



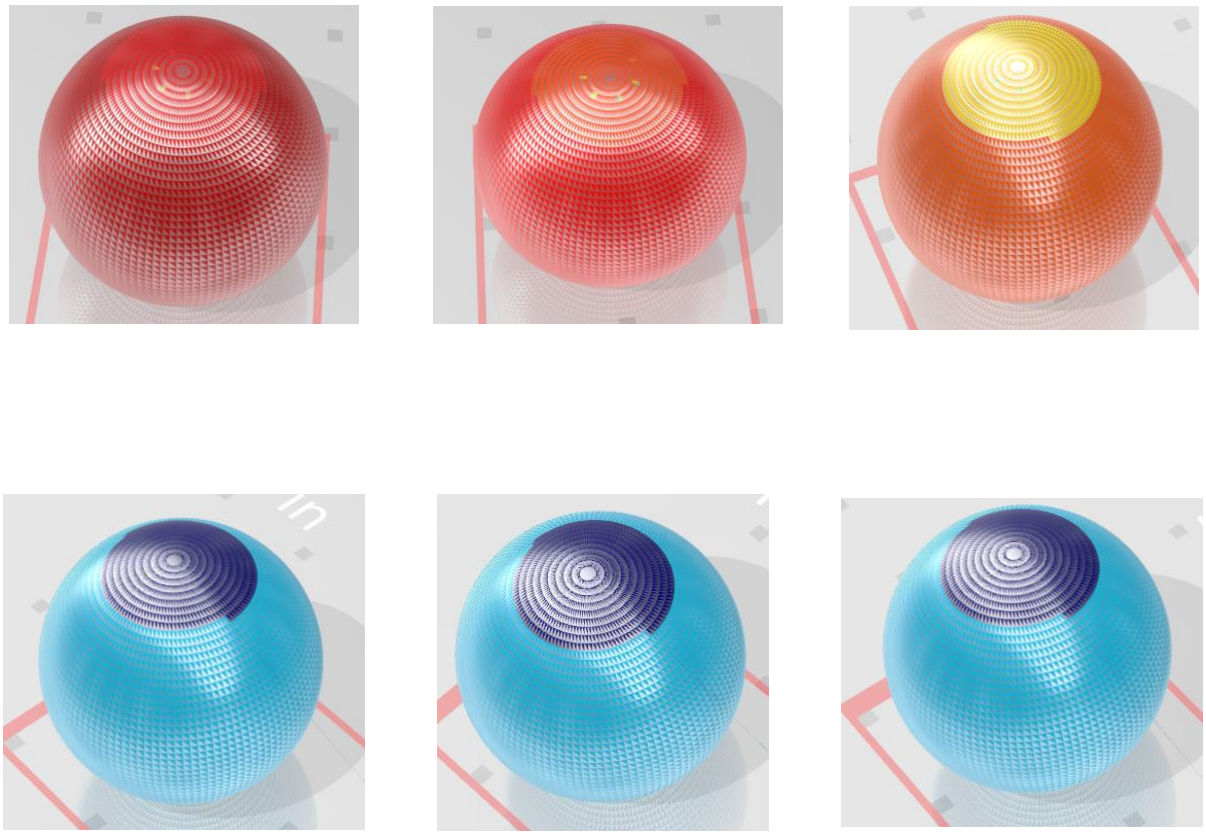
(Fig 1.2 Model of the First Run using Random inputs, Redder tones depict higher temperature values and bluer tones depict lower temperature values. Black is assigned to cells that crossed the threshold of the visualizing software)

In the end, I used the gradients for only two months, one after the winter solstice, January, and one after the summer solstice, August. This was done to ensure that the calculation was run for two scenarios, one where the earth's north is tilted towards the sun and one where the earth's north is tilted away from the sun. As the data set had more points along the upper half of the equator, these two scenarios were chosen assuming that the data for the northern hemisphere was more fine-grained.
The following graphs illustrate the temperature gradients calculated for the mentioned month and year:



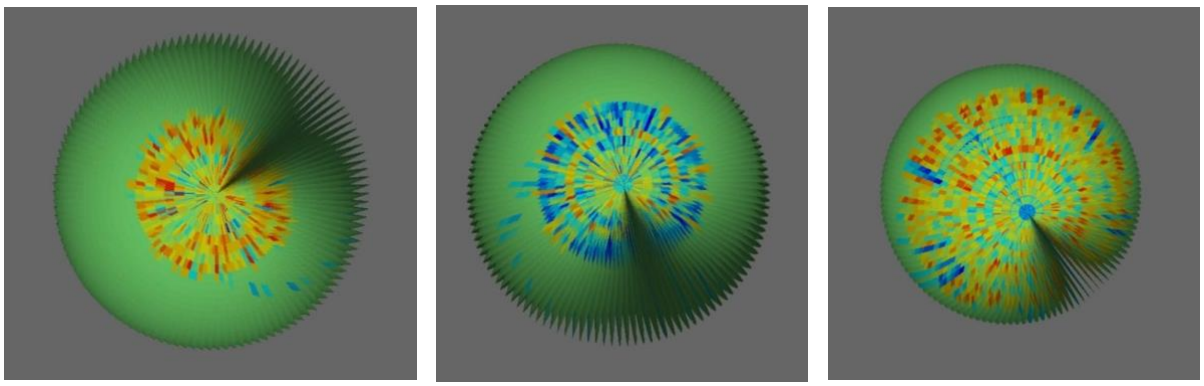


This shows that (as expected) the average gradient north to south is greater than the average gradient east to west, even in terms of angular distances (or actual distances). The datasets used to define the initial state of the Cellular Automaton (the months of January and August) are attached in the Appendix.
In the second run, upon adding the gradients for the longitudes, I got the following result - the automaton stagnated after the 4th time step -



(Fig.1.3 - Fig.1.8, the Visualized output of the automata in successive time steps (1 to 6). The colours change abruptly due to the visualizing software, and the change in values for temperature is more gradual in the output file as seen in the appendix)

This tells us two things that we need to consider while running and creating such a model. We need to differentiate between the two poles so that the equator acts as a sort of mirror along which properties are reflected across both hemispheres.
Using discretized temperature data the following results were obtained -



(Fig 1.9 - Fig. 1.11, the visualized output of the first, second and tenth-time steps when discretized temperature data was used.)

Instead of using the Navier-Stokes from the get-go, I decided to start with the simpler scale-invariant form of the equation suggested by Euler[4],

$$\frac{\delta u}{\delta t} + u - \frac{\delta u}{\delta x} = \frac{-1}{\rho}\frac{\delta P}{\delta x}$$

Where, ρ is the density of the atmosphere in the given cell, u is the velocity of the atmosphere with respect to the earth and P is the pressure. I adapted the above equation into a rule of interaction between the cells to the following[5];

$$u_{i,j} = \frac{\nabla P}{\rho}$$

Where $u_{i,j}$ is the velocity of the cell's atmosphere with i latitude and j longitude. ∇P/ρ will be the site value for each cell calculated from data sets similar to those used for temperature.

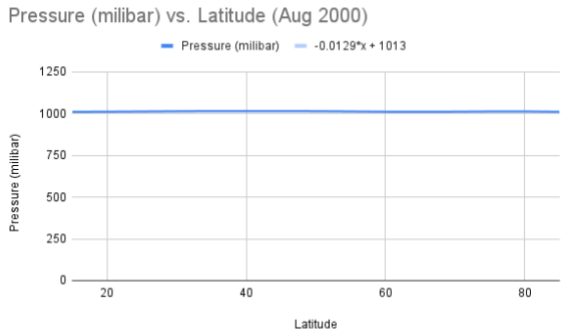
The density of air across lattice points was calculated using the ideal gas equation of state for the atmosphere -

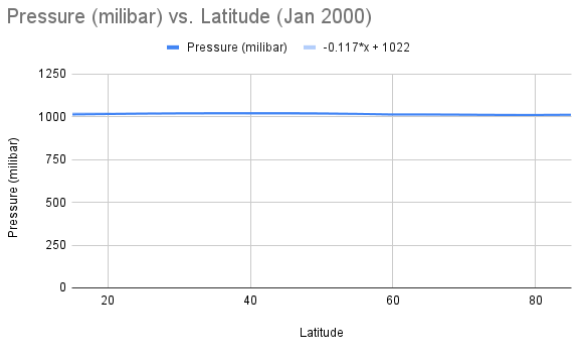
$$\rho = \frac{P}{R * T}$$

where R is the gas constant for the atmosphere at the given point, which is subject to humidity, pollution and altitude. However, due to a lack of sufficient data with respect to humidity, and the problem with factoring humidity only for January and August (which is subject to several other variables in the Brewer-Dobson Cycle), density remained a variable that could not be accounted for.

Velocities serve as an intermediate variable; Pressure may be related to temperature through an equation of state - which would enable prediction of both pressures and temperature together.

Data for Latitudinal Pressure Gradients was taken from (National Center for Atmospheric Research, USA) NCAR's Research Data Archive. The pressure was measured and reported in millibars across a 5x5 Longitudinal Grid from 0° East to 360°E (or 180° W to 0° 180° E) to and a 5x5 Latitudinal Grid from 15°N to 85°N.



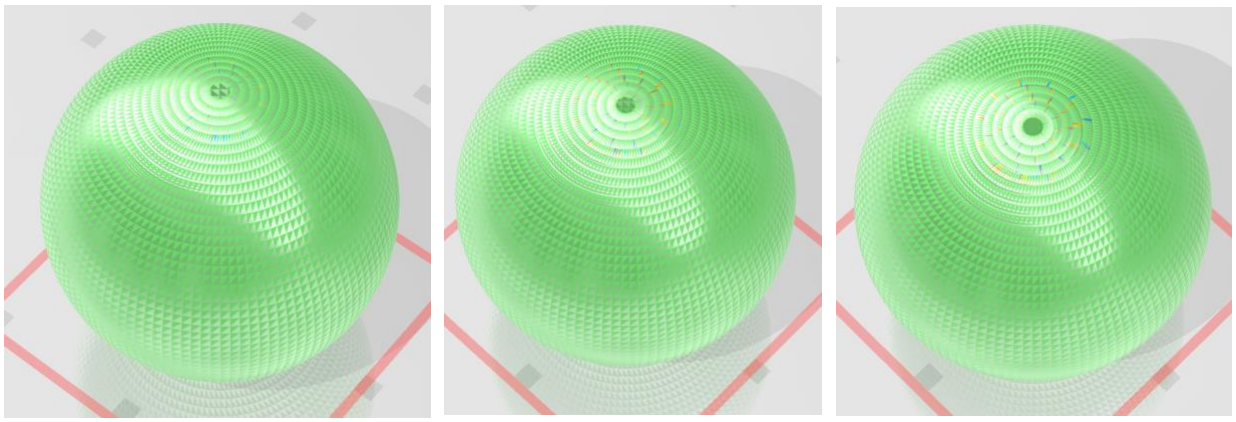


Like with the Temperature data set, Pressure readings taken at sea level across the globe had data for mostly the northern hemisphere. The rule applied to CA for pressure was as follows -

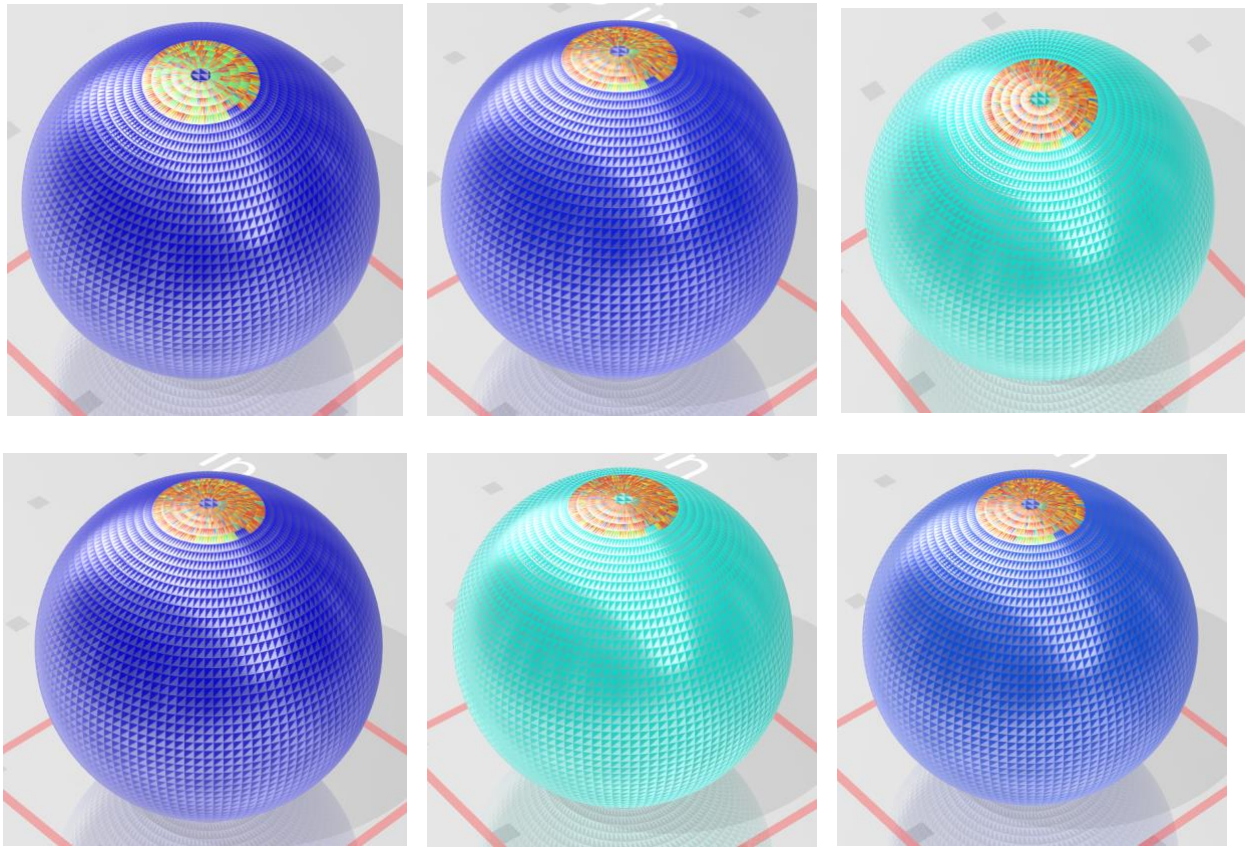
$$P + \frac{dP}{dt} = \frac{dP}{dx} - \frac{dP}{dy}$$

which is not the Navier-Stokes equation, which could not be implemented at this time.

Taking discretized Pressure values (in millibars) and applying the above mentioned CA rule the following result was obtained -



When the automaton was run without the modulo function - the entire lattice homogenised and only extreme values remain,



which is intuitive since the system will try to achieve a state of equilibrium (zero entropy). The modulo function adds an element of stochasticity to the automaton. Since the underlying function used in modulo arithmetic is non-linear, the stochastic behaviour can be expected.

Discussion

The Automata itself was very rudimentary and did not account for several factors that come into play when building a climate model. It is nowhere close to modelling the atmosphere of the Earth. I was not able to build it in a way that could compute more than one variable at a time, so only one variable's change was calculated at a time. It calculated Temperature and Pressure gradients separately, but in reality, both are not exclusive values and the change in one variable plays a significant role in the change in the other.

Another factor that wasn't taken into account is that the atmosphere is heavily influenced by geographical features of the globe, for example, while calculating temperature gradients, the altitude at which the temperature was measured was not taken into consideration before calculating temperature gradients.

However, the automata gave counterintuitive results with respect to temperature. On applying a simple addition rule for temperature, the temperature of the globe ultimately reduced over the time steps before it reached a steady state value. This means that in a roughly spherical system, with temperature gradients similar to that of the Earth's, if the system is not disturbed until it reaches a steady state value, it will settle for lower than average temperatures rather than higher. This was counterintuitive to me because I expected that as time passed, the automata will stagnate at average values around 7.95^{o}C but instead, it stagnated at -1°C. One explanation for this is that I did not account for solar energy flux and night-time radiative loss, both of which affect local temperatures significantly.

The final temperature calculated by the automata at the North Pole is significantly lower than the final temperature calculated for the South Pole. This can be directly related to the data, as the data point for the "South Pole" was calculated and was not a measured statistic from the dataset itself. Thus the initial states of the North went as low as -46°C but the South Pole only went down to -9°C. If we were to assume that the final states of the Northern Hemisphere were more accurate than the Southern Hemisphere, the Automata does deliver on the fact that the pole is colder than the equatorial regions. The difference looks very stark in the visualising software (Figures 1.3 to 1.8) but when we see the actual data output by the automata, the change is pretty gradual (See appendix).

While reading the literature for the project, it was very interesting to note that the climate model across the globe is extremely inconsistent and can draw only one conclusion unanimously. While half the models report an increase in rainfall, the other half report a decrease in precipitation. However, all models report the same increase in temperature with increasing concentrations of CO_2 present in the atmosphere.

References

1. Toffoli, T. Cellular automata as an alternative to (rather than an approximate of) differential equations in modelling physics. Physica D, 10 (1984) 117–127
   (Xin-She Yang and Y. Young, Cellular Automata, PDEs, and Pattern Formation, Chapman, 2005.)
2. Berkeley Earth, 2021, Estimated Global Land-Surface TAVG Based on the complete Berkely Dataset, Berkely Earth,
   http://berkeleyearth.lbl.gov/auto/Global/Complete_TAVG_complete.txt
3. Hurrell, James, Trenberth, Kevin & National Center for Atmospheric Research Staff (Eds). Last modified 03 Apr 2022. "The Climate Data Guide: NCAR Sea Level Pressure." Retrieved from
   https://climatedataguide.ucar.edu/climate-data/ncar-sea-level-pressure .
4. L. D. Landau and E. M. Lifshitz, Fluid Mechanics, Course of Theoretical Physics Volume 6. 2nd Ed. (1987)

External Reading

1. Lin, P. and Fu, Q. (2013), Changes in various branches of the Brewer-Dobson circulation from an ensemble of chemistry-climate models, J. Geophys. Res. Atmos.
2. Li, F., Austin, J., & Wilson, J. (2008). The Strength of the Brewer-Dobson
3. Circulation in a Changing Climate: Coupled Chemistry–Climate Model Simulations, Journal of Climate, 21(1)
4. IPCC, 2019: Climate Change and Land: an IPCC special report on climate change, desertification, land degradation, sustainable land management, food security, and greenhouse gas fluxes in terrestrial ecosystems [P.R. Shukla, J. Skea, E. Calvo Buendia, V. Masson-Delmotte, H.-O. Pörtner, D. C. Roberts, P. Zhai, R. Slade, S. Connors, R. van Diemen, M. Ferrat, E. Haughey, S. Luz, S. Neogi, M. Pathak, J. Petzold, J. Portugal Pereira, P. Vyas, E. Huntley, K. Kissick, M. Belkacemi, J. Malley, (eds.)]
5. P. G. Seybold, L. B. Kier, and C.-K. Cheng, Aurora borealis: stochastic cellular automata simulations of the excited-state dynamics of oxygen atoms. Int. J. Quantum Chem. 1999
6. Yang, Xin-She, and Y. Young. "Cellular automata, PDEs, and pattern formation." arXiv preprint arXiv:1003.1983 (2010).

Acknowledgements

I owe immense gratitude to Dr N. Sukumar for being endlessly patient and understanding with me during the course of the project. His insights and mentoring were the backbone of this project. I would also like to thank my peers who helped me at various points in the process when I felt discouraged and pushed me to see it through to completion.

Appendix

The initial state of the Cellular Automaton for January.

Latitude (in cellular Automata)	Longitude (in cellular Automaton)	Temperature in F	Temperature in C	Latitude	Gradient along latitude (in C)	Longitude	Gradient along longitude (in C)
1	2	56.281	13	-90	56	180	54
2	4	57.79069	14	-87	58	174	53
3	6	59.21056	15	-84	59	168	53
4	8	60.54061	16	-81	61	162	53
5	10	61.78084	17	-78	62	156	52
6	12	62.93125	17	-75	63	150	52
7	14	63.99184	18	-72	64	144	52
8	16	64.96261	18	-69	65	138	51
9	18	65.84356	19	-66	66	132	51
10	20	66.63469	19	-63	67	126	51
11	22	67.336	20	-60	67	120	50
12	24	67.94749	20	-57	68	114	50
13	26	68.46916	20	-54	68	108	50
14	28	68.90101	21	-51	69	102	49
15	30	69.24304	21	-48	69	96	49
16	32	69.49525	21	-45	69	90	49
17	34	69.65764	21	-42	70	84	48
18	36	69.73021	21	-39	70	78	48
19	38	69.71296	21	-36	70	72	48
20	40	69.60589	21	-33	70	66	47
21	42	69.409	21	-30	69	60	47

22	44	69.12229	21	-27	69	54	47
23	46	68.74576	20	-24	69	48	46
24	48	68.27941	20	-21	68	42	46
25	50	67.72324	20	-18	68	36	46
26	52	67.07725	19	-15	67	30	45
27	54	66.34144	19	-12	66	24	45
28	56	65.51581	19	-9	66	18	45
29	58	64.60036	18	-6	65	12	44
30	60	63.59509	18	-3	64	6	44
31	62	61.31509	16	3	61	6	44
32	64	60.04036	16	6	60	12	44
33	66	58.67581	15	9	59	18	45
34	68	57.22144	14	12	57	24	45
35	70	55.67725	13	15	56	30	45
36	72	54.04324	12	18	54	36	46
37	74	52.31941	11	21	52	42	46
38	76	50.50576	10	24	51	48	46
39	78	48.60229	9	27	49	54	47
40	80	46.609	8	30	47	60	47
41	82	44.52589	7	33	45	66	47
42	84	42.35296	6	36	42	72	48
43	86	40.09021	4	39	40	78	48
44	88	37.73764	3	42	38	84	48
45	90	35.29525	2	45	35	90	49
46	92	32.76304	0	48	33	96	49
47	94	30.14101	-1	51	30	102	49
48	96	27.42916	-3	54	27	108	50
49	98	24.62749	-4	57	25	114	50
50	100	21.736	-6	60	22	120	50
51	102	18.75469	-7	63	19	126	51
52	104	15.68356	-9	66	16	132	51

53	106	12.52261	-11	69	13	138	51
54	108	9.27184	-13	72	9	144	52
55	110	5.93125	-14	75	6	150	52
56	112	2.50084	-16	78	3	156	52
57	114	-1.01939	-18	81	-1	162	53
58	116	-4.62944	-20	84	-5	168	53
59	118	-8.32931	-22	87	-8	174	53
60	120	-12.119	-25	90	-12	180	54

The initial state of the Cellular Automaton for August.

Latitude (in cellular Automata)	Longitude (in cellular Automaton)	Temperature in F	Temperature in C	Latitude	Gradient along latitude (in C)	Longitude	Gradient along longitude (in C)
1	2	-51.07	-46	-90	-51	180	69
2	4	-43.5229	-42	-87	-44	174	69
3	6	-36.1936	-38	-84	-36	168	68
4	8	-29.0821	-34	-81	-29	162	68
5	10	-22.1884	-30	-78	-22	156	68
6	12	-15.5125	-26	-75	-16	150	67
7	14	-9.0544	-23	-72	-9	144	67
8	16	-2.8141	-19	-69	-3	138	66
9	18	3.2084	-16	-66	3	132	66
10	20	9.0131	-13	-63	9	126	66
11	22	14.6	-10	-60	15	120	65
12	24	19.9691	-7	-57	20	114	65
13	26	25.1204	-4	-54	25	108	64
14	28	30.0539	-1	-51	30	102	64
15	30	34.7696	2	-48	35	96	64
16	32	39.2675	4	-45	39	90	63
17	34	43.5476	6	-42	44	84	63
18	36	47.6099	9	-39	48	78	63

19	38	51.4544	11	-36	51	72	62
20	40	55.0811	13	-33	55	66	62
21	42	58.49	15	-30	58	60	61
22	44	61.6811	16	-27	62	54	61
23	46	64.6544	18	-24	65	48	61
24	48	67.4099	20	-21	67	42	60
25	50	69.9476	21	-18	70	36	60
26	52	72.2675	22	-15	72	30	59
27	54	74.3696	24	-12	74	24	59
28	56	76.2539	25	-9	76	18	59
29	58	77.9204	26	-6	78	12	58
30	60	79.3691	26	-3	79	6	58
31	62	81.6131	28	3	82	6	58
32	64	82.4084	28	6	82	12	58
33	66	82.9859	28	9	83	18	59
34	68	83.3456	29	12	83	24	59
35	70	83.4875	29	15	83	30	59
36	72	83.4116	29	18	83	36	60
37	74	83.1179	28	21	83	42	60
38	76	82.6064	28	24	83	48	61
39	78	81.8771	28	27	82	54	61
40	80	80.93	27	30	81	60	61
41	82	79.7651	27	33	80	66	62
42	84	78.3824	26	36	78	72	62
43	86	76.7819	25	39	77	78	63
44	88	74.9636	24	42	75	84	63
45	90	72.9275	23	45	73	90	63
46	92	70.6736	21	48	71	96	64
47	94	68.2019	20	51	68	102	64
48	96	65.5124	19	54	66	108	64

49	98	62.6051	17	57	63	114	65
50	100	59.48	15	60	59	120	65
51	102	56.1371	13	63	56	126	66
52	104	52.5764	11	66	53	132	66
53	106	48.7979	9	69	49	138	66
54	108	44.8016	7	72	45	144	67
55	110	40.5875	5	75	41	150	67
56	112	36.1556	2	78	36	156	68
57	114	31.5059	0	81	32	162	68
58	116	26.6384	-3	84	27	168	68
59	118	21.5531	-6	87	22	174	69
60	120	16.25	-9	90	16	180	69

Pressure Gradients Calculated for January and August -

Latitude	Longitude	Jan Pressure (millibar)	Aug Pressure (millibar)
15	24	1014.285	1013.445
20	48	1012.38	1013.26
25	76	1010.475	1013.075
30	96	1008.57	1012.89
35	120	1006.665	1012.705
40	144	1004.76	1012.52
45	168	1002.855	1012.335
50	192	1000.95	1012.15
55	216	999.045	1011.965
60	240	997.14	1011.78
65	264	995.235	1011.595
70	288	993.33	1011.41
75	312	991.425	1011.225
80	336	989.52	1011.04
85	360	987.615	1010.855

Sample output file from the automata and visualizer can be viewed here (modified for interpretation) -
Sample Output File